Battery Rescue Car/Cart

Bay Area Rapid Transit (BART)

Team: 2026-CAP-123 – PDR



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Background



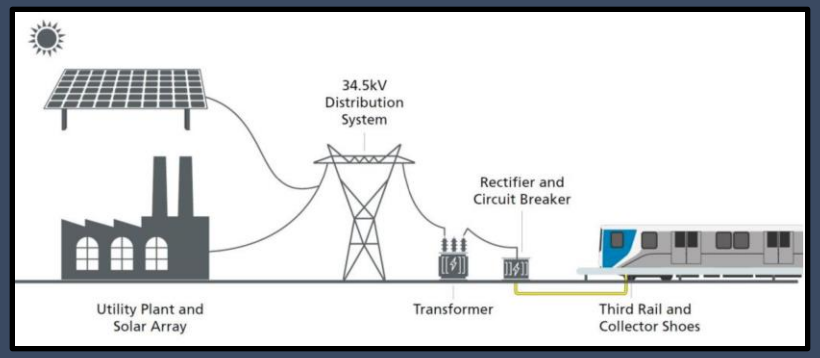
Bay Area Rapid Transit (BART)

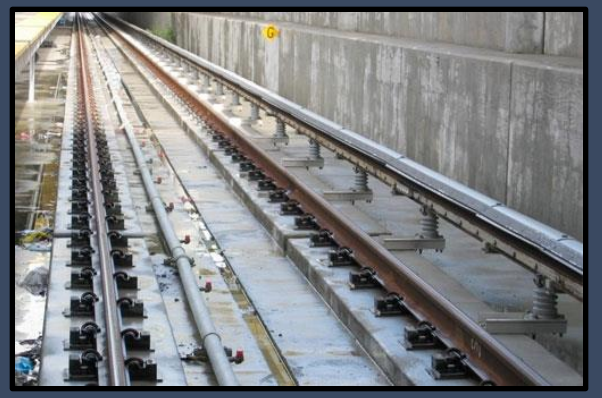
Headquarter Location: Oakland, CA

- BART is a heavy rail public transportation system serving the San Francisco Bay Area
- Operates approximately 131 miles of track and 50 stations across five counties
- BART trains rely on electrical power supplied by a third rail for propulsion
- Alstom is the primary manufacturer for BART's "Fleet of the Future"

Engineering Problem

- BART trains depend on third rail electrical power for propulsion
- Power outages can prevent trains from operating on affected sections of track
  - Difficult to transport personnel and equipment during emergencies and maintenance operations
- BART needs a reliable, independently powered vehicle capable of traveling on both roads and rails





Design Functions and Constraints

Main Function: Safely transport personnel and equipment during power outages

Hard constrains

- Operate independent of third rail power
- Travel on both paved surfaces and railway tracks
- Support the required personnel and equipment load
- Provide safe and reliable operation
- Size and Load
- BART Laws

Soft constrains

- Maximize battery operating time
- Minimize manufacturing and maintenance costs
- Minimize road to rail conversion time
- Easy to operate and maintain

Design Concept 1:

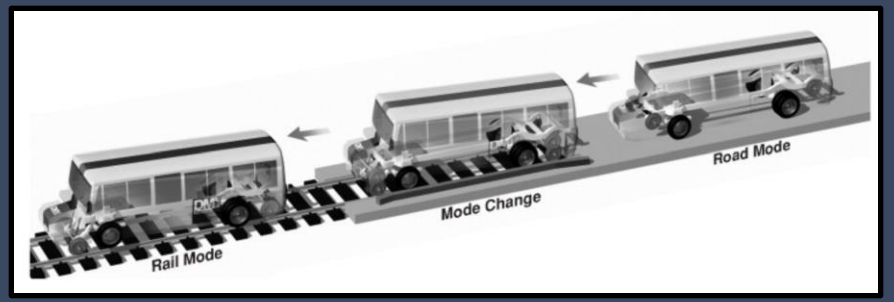
Parallel Guided Wheel Attachment











Main Design

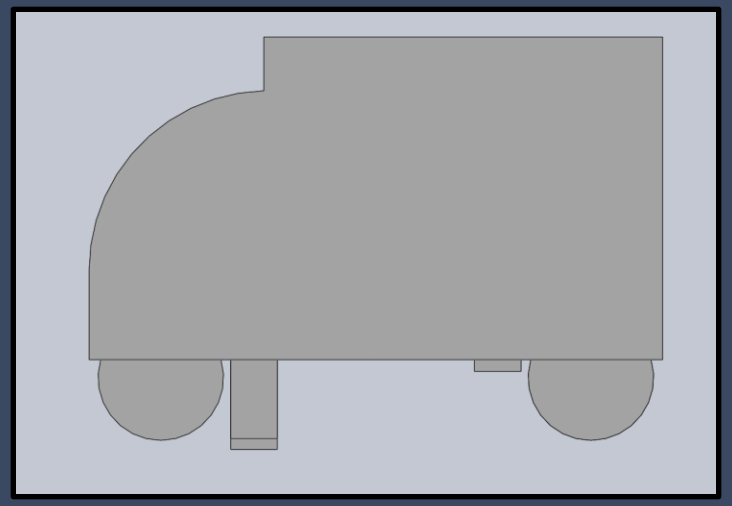
- Hydraulicly powered lever system
- Tires and guide wheels align parallel on track
- Train guide wheels are free spinning
  - Driving power comes direct from car
  - Help lock rotation of car steering

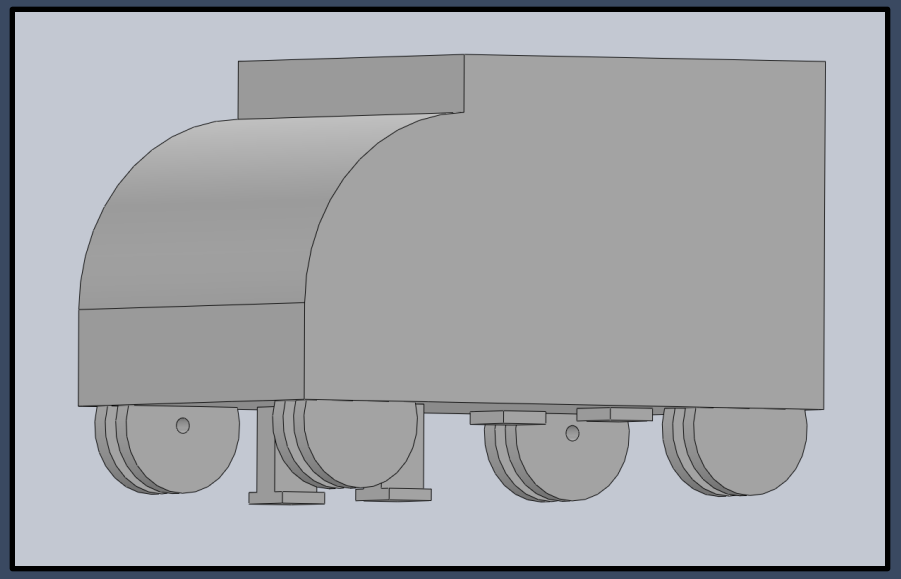
Groups implementing this idea

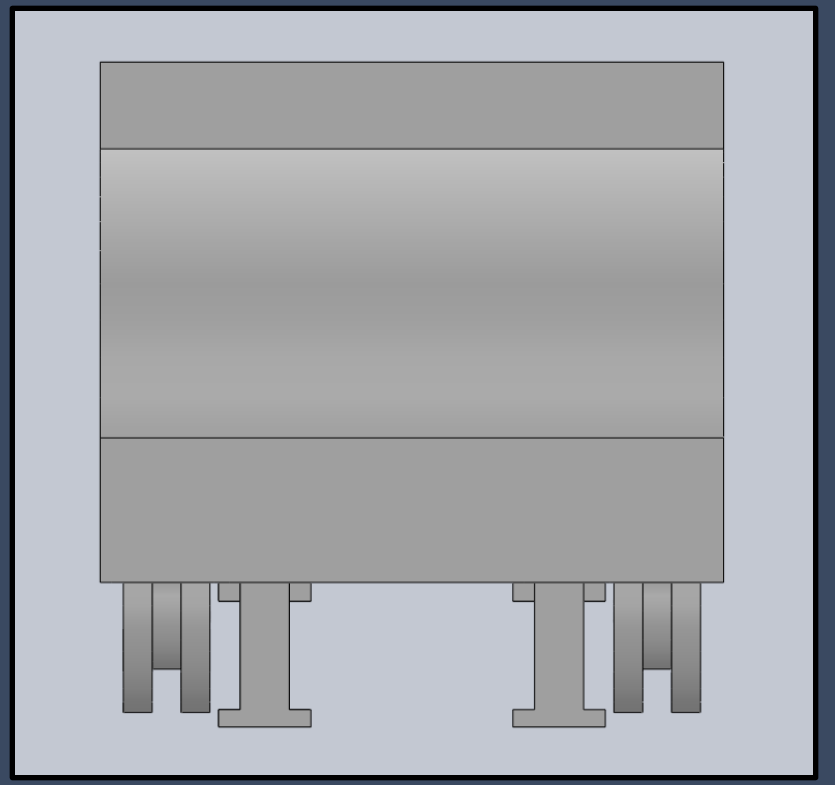
- Custom Truck One Source
- Alke Electric Vehicles
- Dual Mode Vehicle (DMV)

Design Concept 2:

Dually Mated Train Wheel System



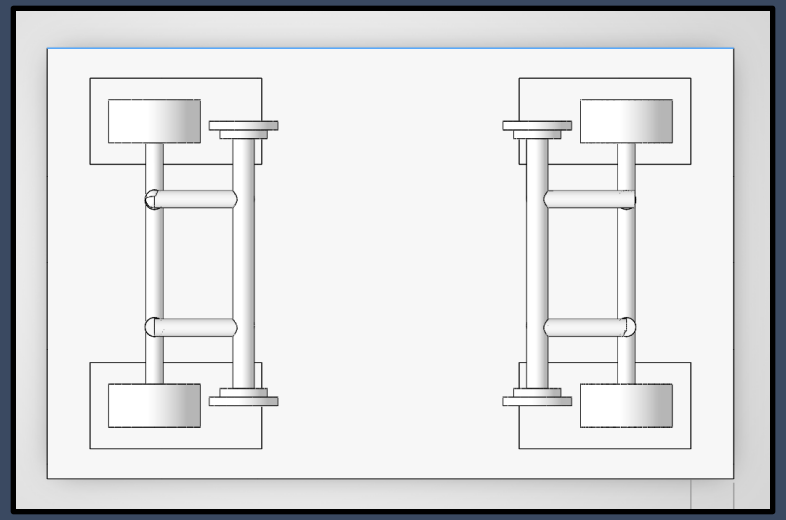


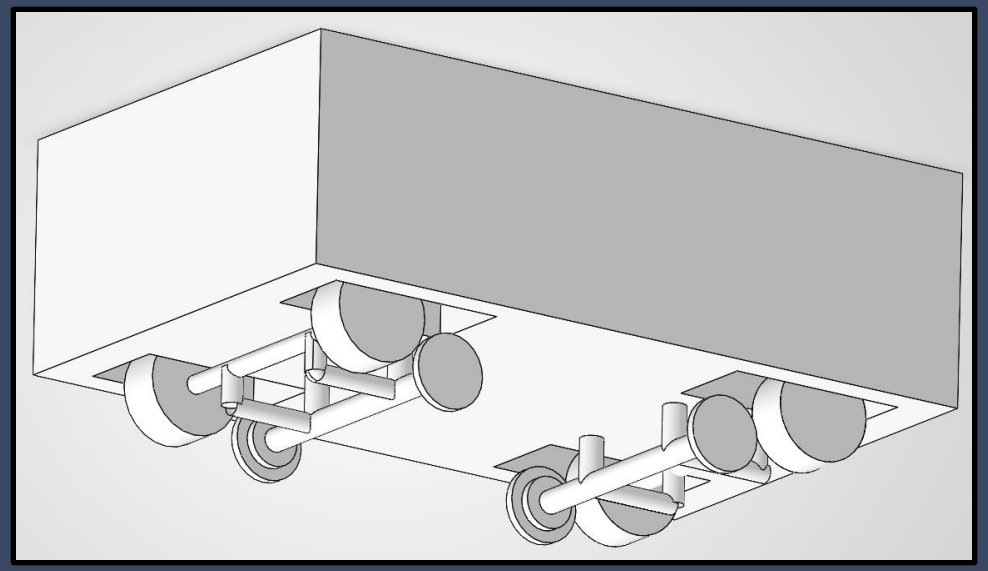


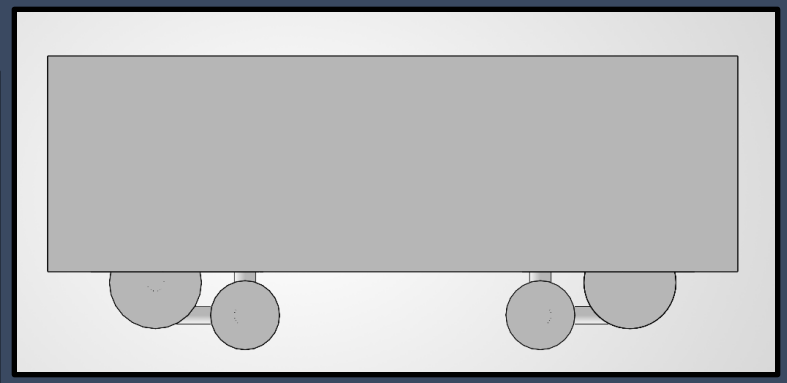
Design Concept 3:

Cavity Latch Axel System

- Railroad wheel attached to the side of the wheel
  - Helps “interchange” between wheels via rotating rod
- System pushes up road wheels to give access to train wheels



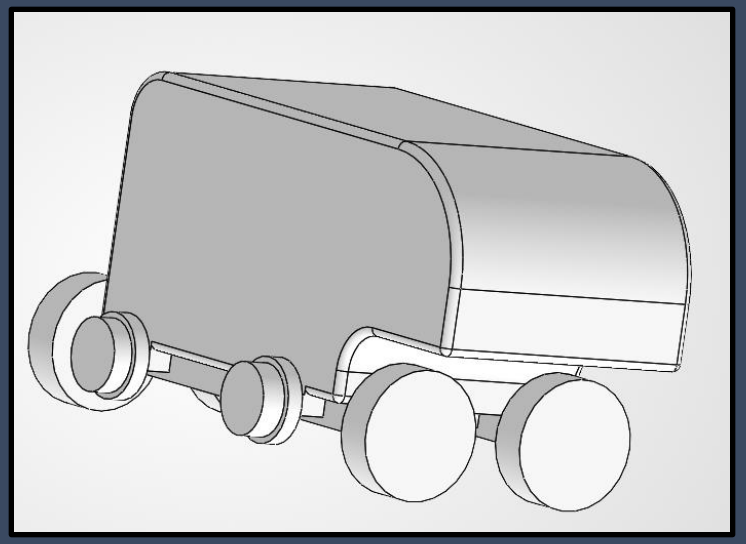


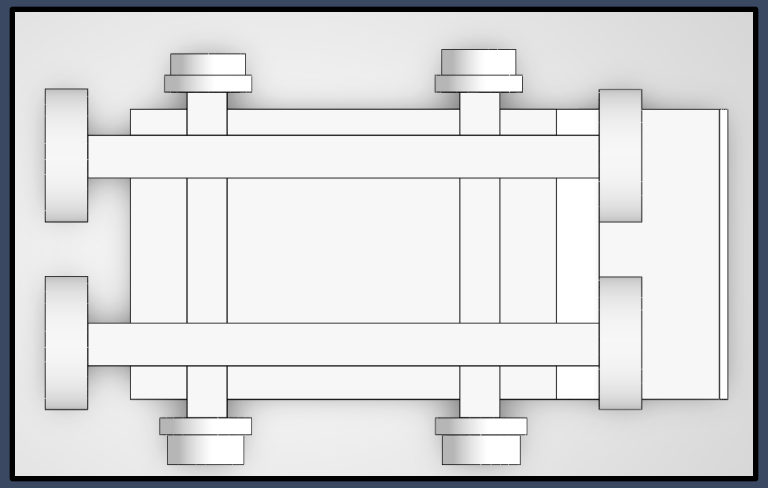


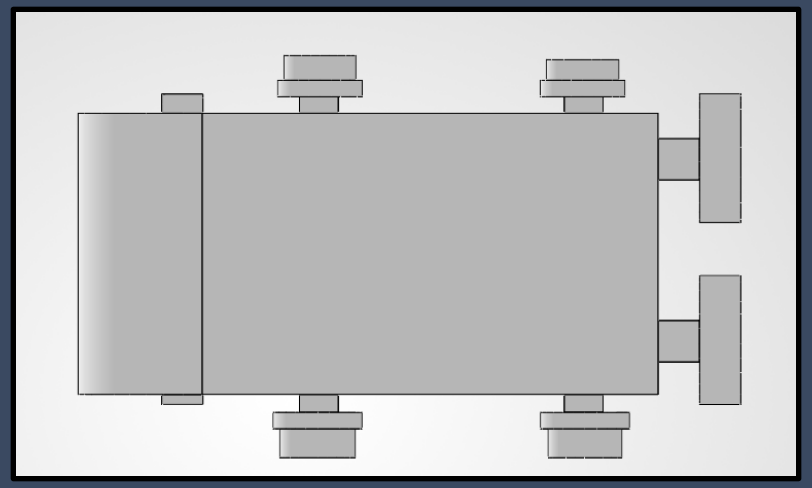
Design Concept 4:

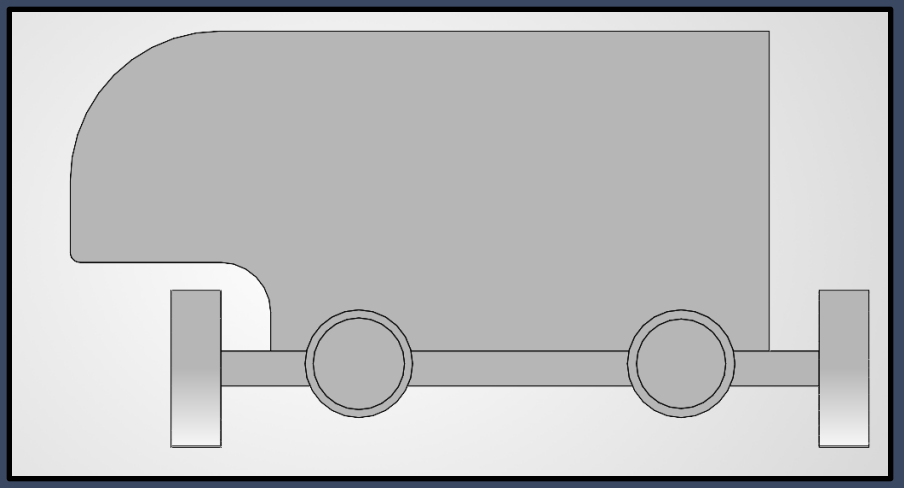
Four-Wheel System

- Two pairs of wheels will be for the road
- Two pairs of wheels will be for the tracks
  - Will have to line up with the tracks







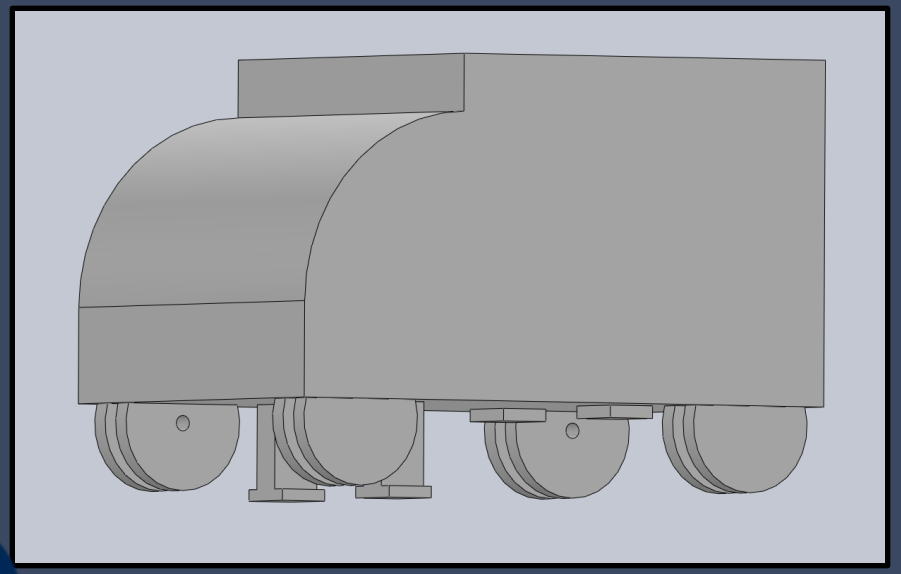


Concept Criteria & Comparison

Score (1-4)	System Integration	Design Feasibility	Road-to-Rail Conversion	Maintenance & Repairability	Ease of Manufacturing	Total Score
Concept 1	4	4	2	3	3	16
Concept 2	3	4	4	3	4	18
Concept 3	3	3	4	3	3	15
Concept 4	2	1	2	2	1	9

Selected Design Concept

Concept 2: **Dually Mated Train Wheel System**



Why We Selected it

- Highest weighted score: 18/20
- **Main advantage**: Hydraulic jack lifts help offer easier access on an off tracks by raising the load off the track
- **Second advantage**: Dual wheel system in front and rear of car/cart to distribute loading from how heavy car/cart will be

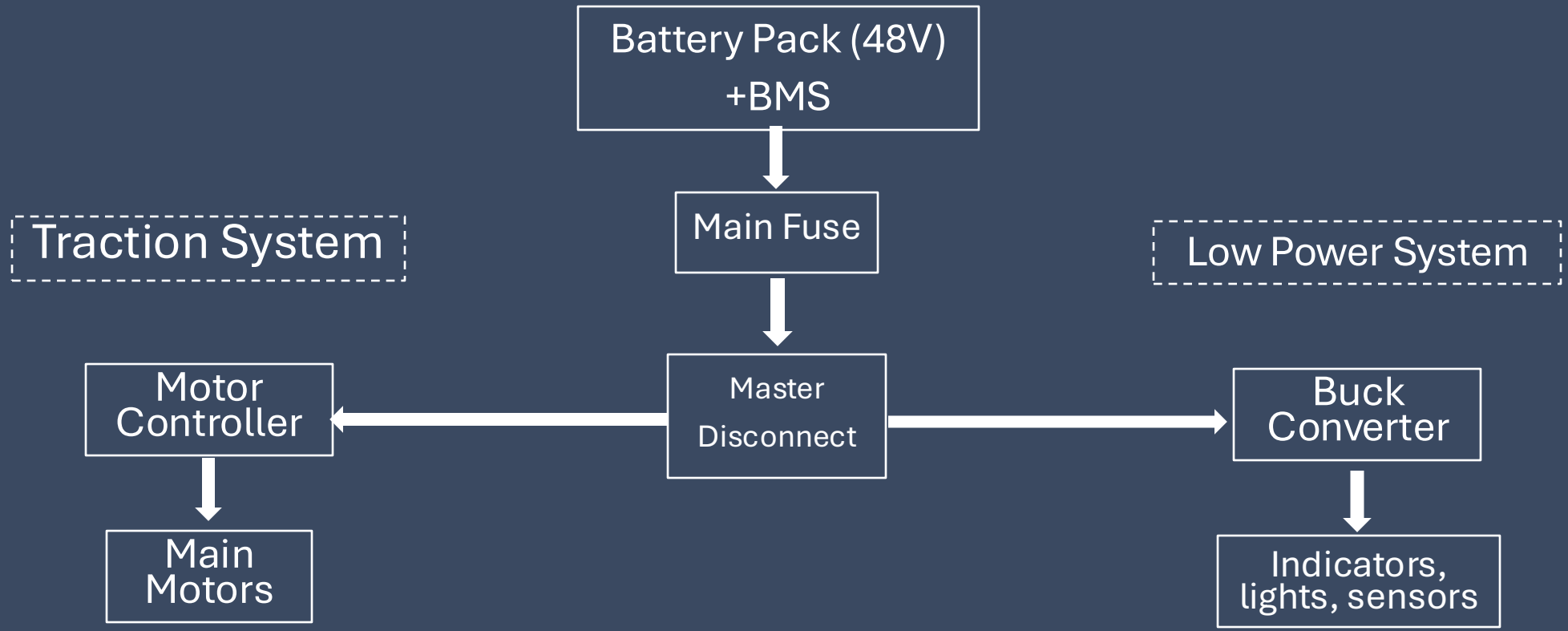
Remaining Challenges

- Implementing cavity space for jack lifts
- Offering lower pressure in car wheels to drive over/off tracks

BART Electrical Laws

- Overcharging protection
- Emergency battery cutoff (fuse, switch)
- Fire detection (temperature, smoke)
- Battery enclosure (noncombustible material)
- Master disconnect (for maintenance)
- Documentation!! (Serviceable and understandable)

Prototype Electrical Diagram



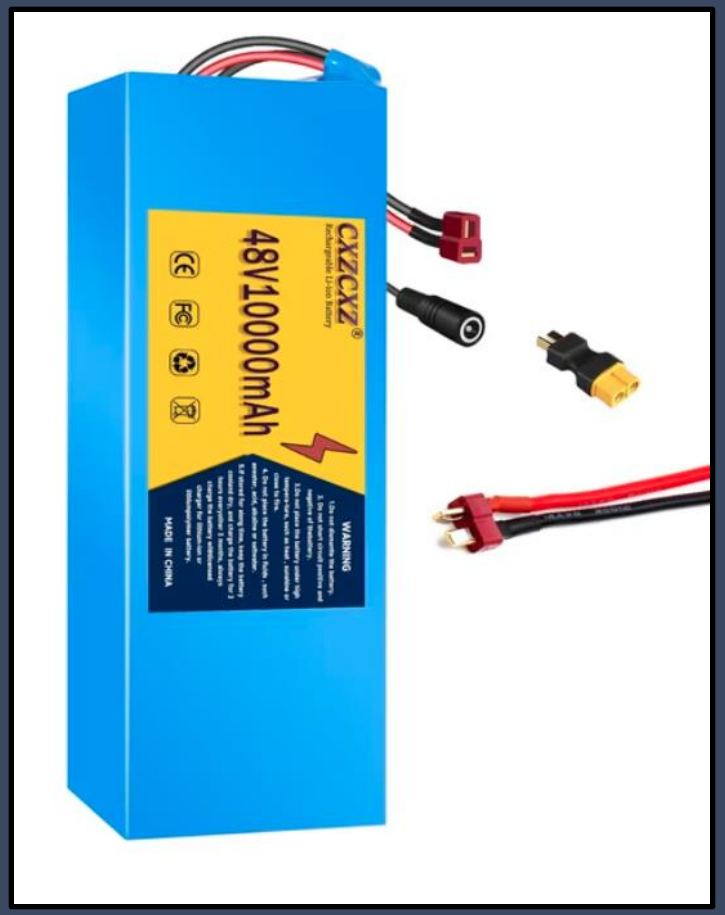
Scooter Battery (Idea 1)

Repurpose a Scooter Battery

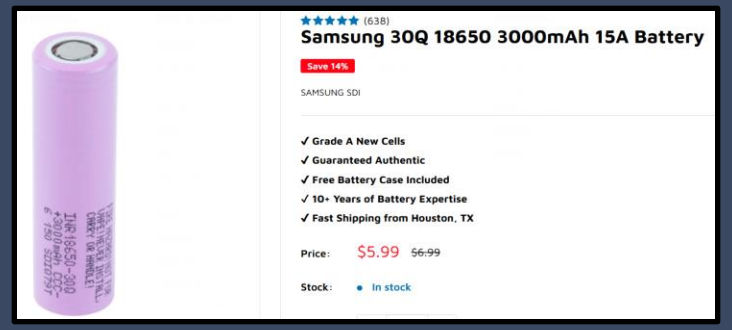
Total Specs

- 48V
- 10 Amp/hr
- 480 Watt/hr

Cost: $117



Make a Battery (Idea 2)



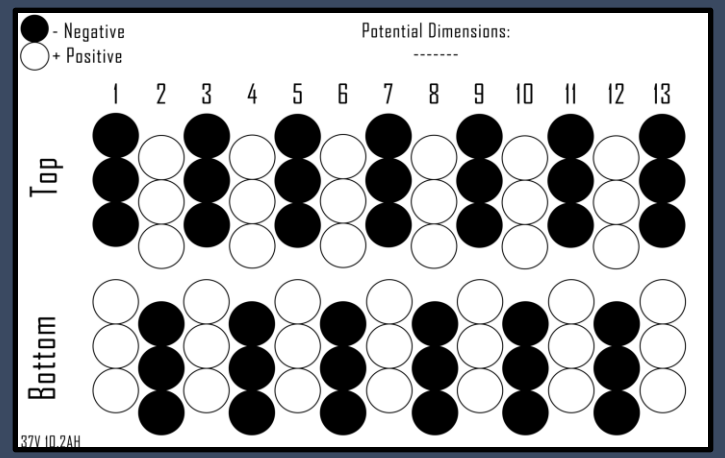
Use Lithium Ion Cells

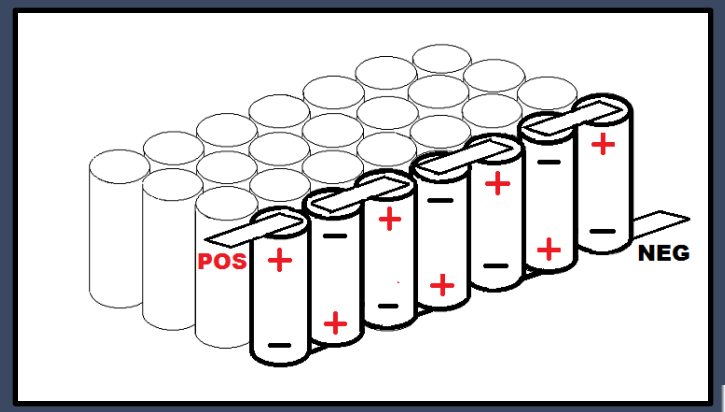
- Each 3.7V, 10 Amp/hr

Total Specs of Full Battery

- This was done for a cost comparison
- 48.1V
- 9 Amp/hr
- 432.9 Watt/hr

Cost: $195 (not including safety components)

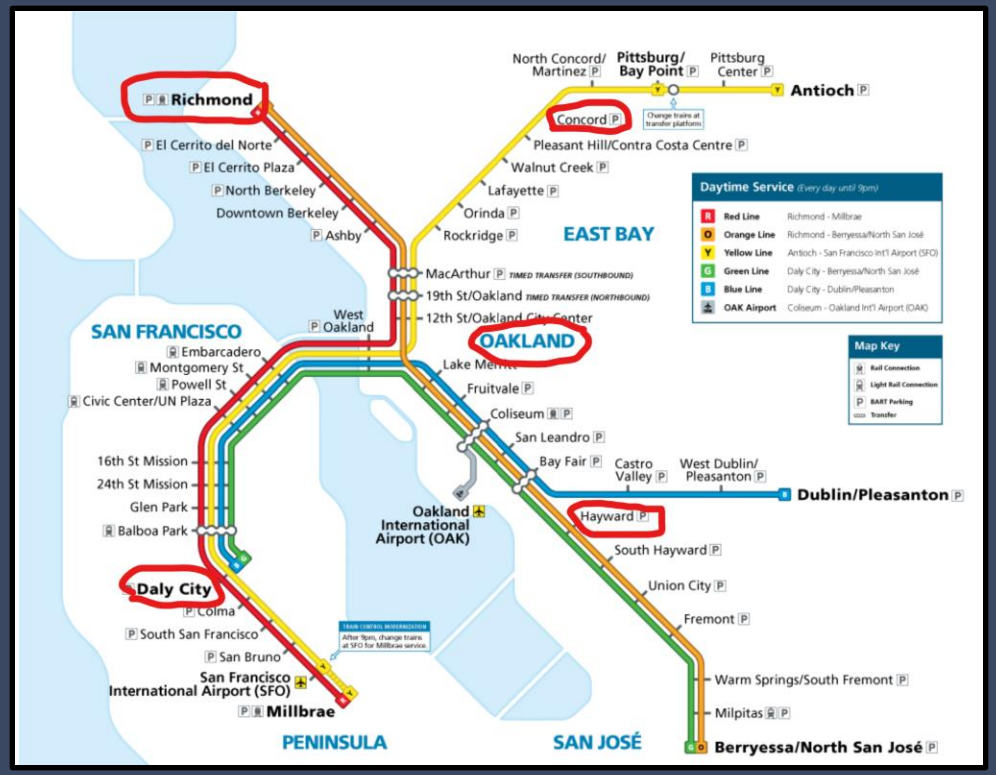




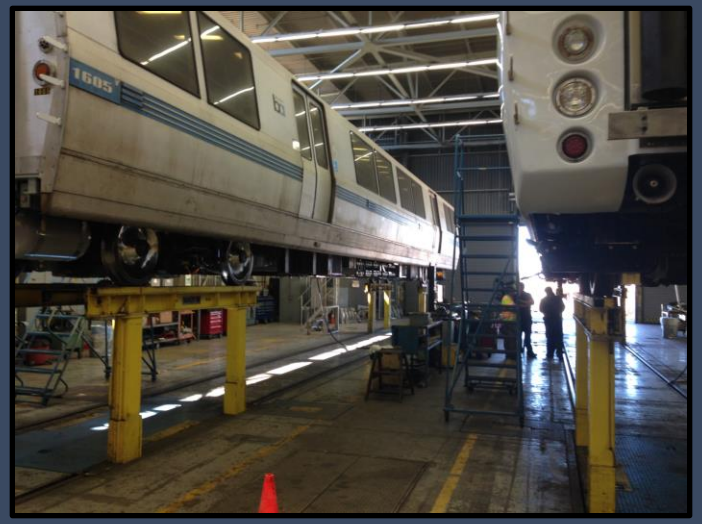
Battery Concept Criteria & Comparison

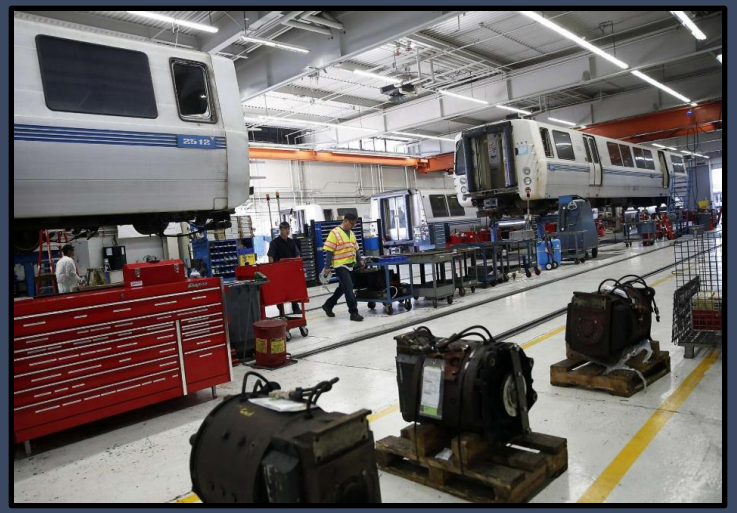
Score (1-4)	Practicality	Safety	Cost	Customizability	Reliability	Total Score
Scooter Battery	4	4	3	3	4	18
Make a Battery	2	2	2	4	3	13

BART Workshop Locations













Project Deliverables and Next steps

Goal: Develop reliable battery powered cart for emergencies

Deliverables

- CAD models of our design including wheel conversion system
- Payload capacity, battery performance, and travel range analysis
- Estimation on fleet size and charging location logistics

Next Steps

- Site visit to Oakland headquarters and maintenance shops
- Finalize the wheel conversion mechanism and component selection
- Develop and test the prototype or proof of concept

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